Greatest common factor

1

a 3 **b** a **c** $3x$ **d** $3b$

2

a $4r + 12 = 4(r + 3)$

b $5v - 30 = 5(v - 6)$

c $18c - 24 = 6(3c - 4)$

d $2t^2 + 2t = 2t(t + 1)$

3

a $6(v + 5)$ **b** $5(9t - 8)$ **c** $-2(s + 5)$ **d** $-2(6s - 5)$

e $y(y + 4)$ **f** $k(k - 5)$ **g** $2u(u - 4)$ **h** $2t(2 + t)$

i $x(42 - x)$ **j** $7(x + 5)$ **k** $6(6 - x)$ **l** $-2(x - 5)$

m $-10(10 + x)$ **n** $-6(3y^2 - 1)$ **o** $13(3x + 4)$ **p** $(t - 3)(8t + 9)$

Grouping with 4 terms

4

a $(5 + v)(a + b)$

b $(y - z)(x - w)$

c $(2y + 3w)(8x - z)$

d $(5x^2 + 4y)(8z - 1)$

e $2(y - 4)(4y - 5)$

f $(4c - d)(c + 9d + 5)$

5

a No **b** Yes **c** Yes **d** No

6

a $(x + 5)(x + 8)$

b $(x - 5)(x + 10)$

7

a $(x + 2)(y - 9)$ **b** $(8 - x)(7 - y)$ **c** $(x + z)(12y + w)$ **d** $(x - 20y)(2 + z)$

e $(5 + x)(10 + y)$ **f** $(12y + w)(x + z)$

8 $(x + 4)(4x + 1)$

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Trinomials

9

a $m + 10$

b $y + 5$

10

a $(x + 1)(x + 5)$

b $(x + 4)(x + 10)$

c $(x - 1)(x - 4)$

d $(x - 3)(x - 10)$

e $(x - 8)(x - 7)$

f $(x + 12)(x - 11)$

g $(x + 15)(x - 6)$

h $(x + 2)(x - 4)$

i $(x + 7)(x - 10)$

j $(x + 2)(x + 3)$

k $(x - 11)(x - 4)$

l $-(x + 1)(x + 12)$

m $-(x - 9)(x - 1)$

11

a $4(x + 6)(x + 8)$

b $-5(x - 4)(x + 2)$

c $3(x - 9)(x + 2)$

d $2x(x + 5)(x + 3)$

12

a $3x^2 + 10x - 8 = (3x - 2)(x + 4)$

b $12x^2 + 25x + 7 = (4x + 7)(3x + 1)$

c $15x^2 - 13x - 6 = (5x - 6)(3x + 1)$

13

a $(7x - 5)(x - 10)$

b $(6x + 7)(x - 1)$

c $(9x - 10)(x - 1)$

d $(7x + 8)(x - 6)$

e $(3x + 2)(2x + 3)$

f $-(4x - 3)(3x + 4)$

g $(9 - 8x)(10 - x)$ or $(8x - 9)(x - 10)$

h $-(7p - 2)(7p + 4)$

14

a $(b + 7)^2$

b $(y - 5)^2$

c $(x + 5)^2$

d $(x + 7)^2$

e $(x - 6)^2$

f $(x - 8)^2$

g $(9x - 4)^2$

h $(9x + 2)^2$

i $(7x - 2)^2$

j $(9 + x)^2$

k $(6 - x)^2$

l $(5 + 6r)^2$

m $7(d + 8)^2$

n $4(x + 5)^2$

o $3(x - 4y)^2$

p $-6(r + 8)^2$

q $-3(x - 2)^2$

r $\left(b - \frac{7}{2}\right)^2$

s $\left(x + \frac{3}{2}\right)^2$

t $\left(x - \frac{3}{2}\right)^2$

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Difference of squares

15

a $p+8$

b $m+6$

16 $x-7$

17 $\left(x^2-10\right)\left(x^2+10\right)$

18 $(ax-b)(ax+b)$

19

a $(x+5)(x-5)$

b $(n-6)(n+6)$

c $(v-7)(v+7)$

d $\left(x-\frac{2}{7}\right)\left(x+\frac{2}{7}\right)$

e $(9-x)(9+x)$

f $(2-u)(2+u)$

g $(2+7y)(2-7y)$

h $(8x-11)(8x+11)$

i $(11m+8)(11m-8)$

j $(9x-10y)(9x+10y)$

k $(11x+7y)(11x-7y)$

l $(4x-5)(4x+5)$

m $6(x-7)(x+7)$

n $3(t+2)(t-2)$

o $5(x-8)(x+8)$

p $4(n+2)(n-2)$

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